

# David Lau

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## EDUCATION

### SUNY Stony Brook University

*Bachelor of Science in Computer Science*

Stony Brook, NY

Aug. 2022 – May 2024

### Rochester Institute of Technology

*Bachelor of Science in Computer Science, transferred*

Rochester, NY

Aug. 2020 – May 2022

## EXPERIENCE

### Software Engineering Intern

*BNY Mellon*

May 2023 – Aug. 2023

*New York, NY*

- Implemented advanced data visualization techniques using Angular and an Oracle relational database, resulting in a 30% increase in user engagement with the financial portal
- Developed Maven libraries that reduced code duplication by 40%, improving the readability and scalability of the code base.
- Revamped the web application into an Angular Progressive Web App (PWA), leading to mobile user adoption and improved accessibility across devices as well as decreased load time by 15%.

### Research and Outreach Intern

*ChangingThePresent*

June 2022 – Aug. 2022

*New York, NY*

- Researched and identified potential strategic partners for the non-profit initiative, resulting in the establishment of partnerships with 12 universities and organization
- Developed and implemented a comprehensive outreach strategy, resulting in a 25% increase in engagement with potential donors and partners.
- Contributed to a non-profit initiative aimed at channeling part of the \$450 billion spent annually on gifts towards charitable donations.

### Tutor

*Stem Nation*

July 2018 – Feb. 2021

*Remote*

- Held weekly feedback sessions with students to understand their challenges, leading to more productive tutoring sessions.
- Analyzed students' progress data to identify areas of improvement and tailored instructional strategies to address their specific needs, leading to an improvement in overall student performance.
- Collaborated with a team of tutors to develop and refine curriculum materials, resulting in a decrease in preparation time for subsequent tutoring sessions.

## PROJECTS

### Geodudes | *React, Redux, Leaflet, Node.js, Express, MongoDB*

- Developed GeoDudes, a comprehensive mapping platform, utilizing MERN stack, and redux for efficient state management.
- Integrated an advanced heatmap feature to visualize geospatial data density and also intelligently cluster data points, displaying the count within each cluster for a concise representation of large datasets across the globe.
- Automated deployment to render using GitHub Actions that initiates comprehensive testing using Cypress, Supertest, and Jest, ensuring automatic deployments upon successful push to the main branch.

### Stack Overflow Clone | *MongoDB, Express.js, React.js, Node.js*

- Developed and deployed a fully functional clone of Stack Overflow using the MERN stack.
- Implemented user creation, login, and secure password hashing to ensure a seamless and secure user experience.
- Utilized Express sessions for efficient session management, resulting in improved overall application performance.

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, C, JavaScript, HTML/CSS

**Tools & Technologies:** Git (GitHub, Gitlab), Jira, Trello, Maven, Spring, MongoDB, SQL, VS Code, IntelliJ

**Languages:** English, Cantonese

**Transferable Skills:** Time management, Problem Solving, Communication, Adaptability